



Specialized Topics of Theoretical Physics

Majorana Modes in the Machine – Simulating Topological Phases with Quantum Circuits and AI

NISQ Noise on the Diagnostic

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1 Introduction

The 1D Kitaev chain hosts unpaired Majorana zero modes at its ends in the topological phase [1]. In the preceding part of the project, a parity-constrained variational quantum eigensolver (VQE) prepared the finite-chain ground state and measured the non-local edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$. That ideal result does not yet answer the experimentally relevant question: does the diagnostic remain visible after imperfect gates, decoherence, readout errors, and a noisy variational loop?

This report covers Part 5, *NISQ Noise on the Diagnostic*. It isolates noise at three different levels—a frozen state, the gates of the preparation circuit, and the variational parameters—and then removes two simplifying assumptions by optimizing the noisy cost and using a recorded backend calibration. The emphasis is quantitative: the same $L = 4$ circuit and edge-string observable are retained while only the noise model or ansatz depth is changed.

The noiseless construction of the VQE and string-measurement circuit belongs to Zhengyi Liu's Part 4. The symmetry-based taxonomy of physical Kitaev-chain errors and the scaling to large L belong to Edgar Harutyunyan's Part 6; here T_1/T_2 enter only as device-level channels acting on the simulated circuit.

2 Noise model and diagnostic

2.1 Gate-level quantum channels

An ideal gate g acts on a density matrix as $\mathcal{U}_g(\rho) = U_g \rho U_g^\dagger$. In the circuit-level model the ideal gate is followed immediately by a completely positive trace-preserving channel,

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \quad \mathcal{E}_g(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I. \quad (2.1)$$

For a circuit containing M gates, the output is the ordered composition $\rho_{\text{out}} = (\mathcal{E}_{g_M} \circ \mathcal{U}_{g_M}) \circ \cdots \circ (\mathcal{E}_{g_1} \circ \mathcal{U}_{g_1})(\rho_0)$. Noise therefore accumulates with the actual transpiled gate count rather than being applied once to the final state. Qiskit Aer implements this construction with a density-matrix simulator and a `NoiseModel` [3].

The controlled baseline attaches a two-qubit depolarizing channel after every `cx`,

$$\mathcal{E}_p^{\text{dep}}(\rho) = (1 - p)\rho + p \frac{\text{Tr} \rho}{2^n} I. \quad (2.2)$$

A second model uses gate-duration-dependent thermal relaxation. For a gate of duration τ , amplitude loss has probability $p_{T_1} = 1 - e^{-\tau/T_1}$, while T_2 controls the decay of off-diagonal coherences. The two mechanisms differ operationally: relaxation may move the prepared state out of its target parity sector, whereas dephasing directly suppresses the endpoint coherence needed by the edge string. These are channel mechanics only; their classification as physical errors of the Kitaev chain is deferred to Part 6.

2.2 Three locations for error

The study separates effects that can otherwise be confused. The Week-8 *frozen-state* test takes the validated state and applies readout or a single post-preparation channel. Symmetric assignment error ϵ

attenuates a weight- L Pauli string as

$$\langle O_{\text{edge}} \rangle_{\text{readout}} = (1 - 2\epsilon)^L \langle O_{\text{edge}} \rangle, \quad (2.3)$$

with an additional statistical uncertainty of order $N_{\text{shots}}^{-1/2}$. The *circuit-level* test instead applies a channel after every entangling gate. Finally, a coherent calibration error replaces θ^* by $\theta^* + \delta\theta$ but keeps the state pure; an incoherent channel generally lowers $\text{Tr} \rho^2$. Purity therefore distinguishes a unitary miscalibration from genuine mixing even when both reduce $|\langle O_{\text{edge}} \rangle|$.

2.3 Expressibility versus accumulated noise

For the linear-entanglement EfficientSU2 ansatz, r repetitions on L qubits contain $r(L - 1)$ CNOTs. A useful approximation to the measured signal is

$$|\langle O_{\text{edge}} \rangle|_{\text{meas}}(r) \simeq |\langle O_{\text{edge}} \rangle|_{\text{ideal}}(r) (1 - p_{\text{cx}})^{r(L-1)}. \quad (2.4)$$

The first factor can have an expressibility threshold; the second decreases at every added layer. The relevant optimum is consequently the shallowest depth that can represent the target state, not the deepest affordable circuit.

Putting noise inside the VQE changes the objective itself. With parity $P = \prod_j Z_j$, the cost evaluated on the noisy density matrix is

$$C_{\text{noisy}}(\theta) = \text{Tr}[H\rho_\theta] + \lambda(1 - \text{Tr}[P\rho_\theta])^2, \quad \rho_\theta = \mathcal{E}\left(U(\theta)|0\rangle\langle 0|U^\dagger(\theta)\right). \quad (2.5)$$

A lower value of Eq. (2.5), or even a larger noisy edge string, need not imply a higher-fidelity ground state. The re-optimized angles must therefore be checked in a noiseless circuit against the exact ground state.

3 Methods and Results

3.1 Circuit-level implementation and verification

All circuit results use $L = 4$, $t = \Delta = 1$, the same parity-constrained VQE and edge string as Part 4, and a linear RY+cx EfficientSU2 ansatz [2]. Circuits are transpiled at optimization level zero to the fixed basis [rz,sx,cx], ensuring that an error registered on cx fires after every entangling instruction. The state is saved as a density matrix and the diagnostic is evaluated as $\text{Tr}(O_{\text{edge}}\rho)$.

The integration was verified independently. A reference routine evolves the density matrix gate by gate with `Operator` and explicitly applies a depolarizing `SuperOp` after each CNOT; it does not use Aer's noise machinery. Across 41 values of μ/t , the reference and Aer results agree at approximately 10^{-15} . Figure 1 also compares the deliberately strong $p_{\text{cx}} = 0.10$ depolarizing channel with thermal relaxation at $T_1 = 80 \mu\text{s}$, $T_2 = 60 \mu\text{s}$ and $\tau_{\text{cx}} = 300 \text{ ns}$. The associated one-gate relaxation probability is only $1 - e^{-\tau/T_1} \simeq 0.0037$, so the thermal curve remains closer to the ideal one.

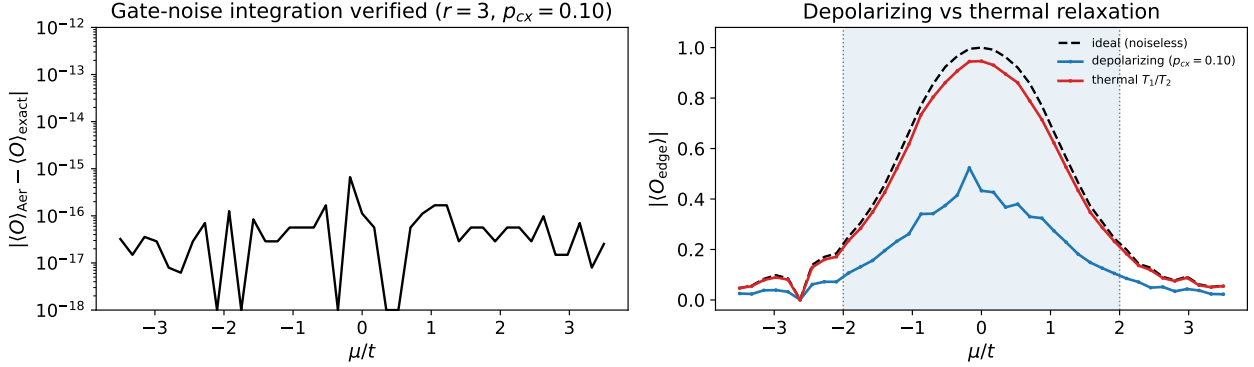


Figure 1: Left: the Aer NoiseModel edge string and the independent exact gate-by-gate reference agree to $\sim 10^{-15}$ across the μ sweep. Right: depolarizing versus thermal T_1/T_2 noise at matched gate placement.

3.2 The optimal depth is an expressibility threshold

At the topological sweet spot $\mu = 0$, the noiseless edge string is only $\simeq 0.004$ for $r = 1$ but jumps to approximately one at $r = 2$ and remains saturated for deeper circuits. Thus the failure at $r = 1$ is under-expressibility, not an optimizer accident. With $p_{cx} = 0.05$, the signal peaks at $r^* = 2$; every further repetition adds three noisy CNOTs without an ideal-state gain. Figure 2 makes the two factors in Eq. (2.4) visible separately.

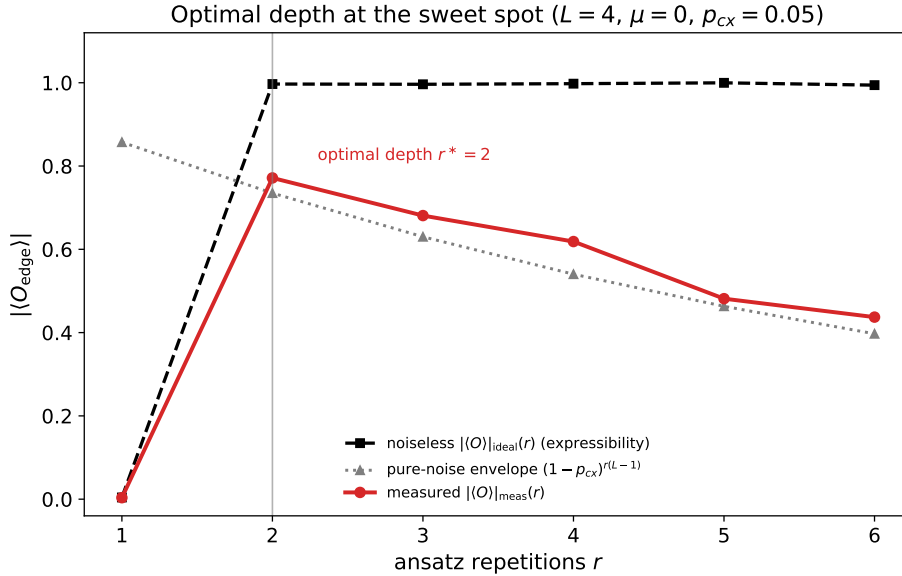


Figure 2: For $L = 4$, the noiseless edge string has an expressibility threshold: $r = 1$ fails, $r = 2$ succeeds, and deeper circuits mainly accumulate CNOT noise.

Parameter perturbations provide a complementary diagnostic. Gaussian $\delta\theta$ rotates the ideal state away from the optimum while its purity stays at one. Per-CNOT depolarization suppresses the same edge string while the purity falls. The two errors should therefore not be summarized by a single “effective gate error”: coherent drift is in principle recalibratable, whereas incoherent loss cannot be undone by retuning one angle vector.

3.3 Noisy optimization and backend calibration

The noisy-cost experiment uses $L = 4$, $r = 3$, $\mu = 0$, $\lambda = 0.1$ and COBYLA, warm-started from the noiseless optimum with additional random starts. Depolarizing noise is unital and produces almost no exploitable displacement of the optimum. Amplitude damping is non-unital: re-optimization can increase the *measured* edge string while moving the underlying noiseless circuit away from the exact ground state. Selected values from the reproducible seed-7 sweep are listed in Table 1.

channel	strength	$ \langle O_{\text{edge}} \rangle $ noisy		fidelity to exact GS	
		fixed θ^*	noise-aware	fixed θ^*	noise-aware
depolarizing	0.08	0.502	0.501	0.998	1.000
amplitude damping	0.08	0.422	0.565	0.998	0.994
amplitude damping	0.12	0.266	0.337	0.998	0.966

Table 1: Best-case versus noise-aware optimization at $L = 4$, $r = 3$, $\mu = 0$. Under amplitude damping the measured edge string rises while the noiseless fidelity to the exact ground state falls at strong damping.

Finally, a `NoiseModel.from_backend` calculation replaces the uniform toy channel with the recorded FakeManilaV2 calibration. Its heterogeneous CNOT error rates are about 0.009–0.014. At $\mu = 0$ the ideal, backend-calibrated, and uniform $p_{\text{cx}} = 0.05$ edge strings are 0.99, 0.87, and 0.67, respectively. The uniform model is therefore a conservative caricature rather than a calibrated prediction. This comparison is still mildly optimistic: the backend noise model contains readout errors, but they do not fire because the simulation reads $\text{Tr}(O_{\text{edge}}\rho)$ directly without measurement instructions. The calculation uses a recorded calibration snapshot, not a live hardware run.

Together these tests show that realistic noise primarily reduces the contrast of the non-local diagnostic. Circuit depth should be chosen by noiseless expressibility validation followed by noisy evaluation, and any noise-aware optimization must report a state-quality check rather than only the observable it was able to improve. These conclusions concern the $L = 4$ NISQ circuit; their large- L consequences are treated separately in Part 6.

4 On the AI collaboration

A large-language model was used to organize this first draft, condense the Block-4 notes, and check that the numerical claims match `block4.py`. It also helped separate frozen-state, circuit-level, parameter-level, and optimizer-level noise so that results from different experiments were not silently combined. The main correction during the AI-assisted review was to reject the tempting statement that noise-aware VQE “recovers” the topological signal: the independent noiseless-fidelity check shows that a non-unital channel can improve the displayed edge string while producing a worse state. All AI-drafted prose remains marked in purple and should be revised by the author before submission.

References

- [1] A. Yu. Kitaev. Unpaired Majorana fermions in quantum wires. *Physics-Uspekhi*, 44(10S):131–136, 2001. doi: 10.1070/1063-7869/44/10S/S29.
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- [3] Qiskit contributors. Qiskit: An open-source framework for quantum computing. 2024. doi: 10.5281/zenodo.2573505.